Isometry Pursuit: Convex Recovery of Orthonormal Column Submatrices

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Abstract

Isometry pursuit is a convex algorithm for identifying orthonormal column-submatrices of wide matrices. It consists of a vector normalization followed by multitask basis pursuit. Applied to Jacobians of putative coordinate functions, it helps identity locally isometric embeddings from within interpretable dictionaries. We provide theoretical and experimental results justifying this method, including a proof with realistic assumptions that such isometric submatrices, should they exist, are contained within the obtained support. For problems involving coordinate selection and diversification, it offers a synergistic alternative to greedy and brute force search.

1 Introduction

Many real-world problems may be abstracted as selecting a subset of the columns of a matrix representing stochastic observations or analytically exact data. This paper focuses on a simple such problem. Given a rank D matrix $X \in \mathbb{R}^{D\times P}$ with $P > D$, select a square submatrix $X_{\cdot S}$ where subset $S \subset [P]$ satisfies $|S| = D$ that is as isometric as possible.

This problem arises in interpretable learning specifically because while the coordinate functions of a given feature space may have no intrinsic meaning, it is sometimes possible to generate a dictionary of interpretable features for consideration as potential parametrizing coordinates. When this is the case, selection of candidate interpretable features as coordinates can take the above form. While implementations vary across data and algorithmic domains, identification of such coordinates generally aids mechanistic understanding, generative control, and statistical efficiency.

This paper shows that an adapted version of the algorithm in Koelle et al. (2024) leads to a convex alternative to the greedy approaches such as those in Cai and Wang (2011); Chen and Meila (2019); Kohli et al. (2021); Jones et al. (2007) for finding isometries. This algorithm - IsometryPursuit - is multitask basis pursuit applied to an appropriately normalized X selects orthonormal submatrices. Given vectors in $\mathbb{R}^D$, the normalization log-symmetrizes length and favors those closer to unit length, while basis pursuit favors those which are orthogonal. Our theoretical results formalize this intuition, and show the usefulness of IsometryPursuit as a trimming procedure prior to brute force search for diversification and interpretable coordinate selection.

2 Background

Our algorithm is motivated by spectral and convex analysis.

2.1 Problem

Our goal is, given a matrix $X \in \mathbb{R}^{D\times P}$, to select a subset $S \subset [P]$ with $|S| = D$ such that $X_{\cdot S}$ is as orthonormal as possible in a computationally efficient way. To this end, we define a ground truth loss function that measures orthonormalness, and then introduce a surrogate loss function that convexifies the problem so that it may be efficiently solved. Figure 1a and 1b provide a visual depiction of this problem when $D = 2$ and $P = 30$.

2.2 Interpretability and isometry

Our motivating example is the selection of data representations from within sets of putative coordinates: the columns of a provided wide matrix. Compared with Sparse PCA (d'Aspremont et al., 2007; Dey et al., 2017; Bertsimas and Kitane, 2022; Bertsimas et al., 2022), we seek a low-dimensional representation from the set of these column vectors rather than their span.

This method applies to interpretability, for which parsimony is at a premium. Interpretability arises through comparison of data with what is known to be important in the domain of the problem. This knowledge often takes the form of a functional dictionary. Evaluation of independence of dictionary features arises in numerous scenarios (Chen and Meilă, 2019; Koelle et al., 2022; He et al., 2023). The requirement that dictionary features be full rank has been called functional independence (Koelle et al., 2022) or feature decomposability (Templeton et al., 2024), with connection between dictionary rank and independence via the implicit function theorem. Besides independence, the metric properties of such dictionary elements are of natural interest. This is formalized through the notion of differential.

Definition 1 *The **differential** of a smooth map $\phi : \mathcal{M} \to \mathcal{N}$ between D dimensional manifolds $\mathcal{M} \subseteq \mathbb{R}^B$ and $\mathcal{N} \subseteq \mathbb{R}^P$ is a map in tangent bases $x_1 \dots x_D$ of $T_\xi \mathcal{M}$ and $y_1 \dots y_D$ of $T_{\phi(\xi)}\mathcal{N}$ consisting of entries*

$$D\phi(\xi) = \begin{bmatrix} \frac{\partial \phi_1}{\partial x_1}(\xi) & \dots & \frac{\partial \phi_1}{\partial x_D}(\xi) \\ \vdots & & \vdots \\ \frac{\partial \phi_D}{\partial x_1}(\xi) & \dots & \frac{\partial \phi_D}{\partial x_D}(\xi) \end{bmatrix}. \tag{1}$$

It is not always necessary to explicitly estimate tangent spaces when applying this definition. The most commonly encountered manifolds are vector spaces for which the tangent spaces are trivial. This is the case for full-rank tabular data, for which isometry has a natural interpretation as a type of diversification, and often for the latent spaces of deep learning models. In this case, $B = D$.

Definition 2 *A map ϕ between D dimensional submanifolds with inherited Euclidean metric $\mathcal{M} \subseteq R^B$ and $\mathcal{N} \subseteq R^P$ ϕ is an **isometry at a point** $\xi \in \mathcal{M}$ if*

$$D\phi(\xi)^T D\phi(\xi) = I_D. \tag{2}$$

That is, ϕ is an isometry at ξ if $D\phi(\xi)$ is orthonormal.

The applications of pointwise isometry are themselves manifold. Pointwise isometric embeddings faithfully preserve high-dimensional geometry. For example, Local Tangent Space Alignment (Zhang and Zha, 2004), Multidimensional Scaling (Chen and Buja, 2009) and Isomap (Tenenbaum et al., 2000) non-parametrically estimate embeddings that are as isometric as possible. Another approach stitches together pointwise isometries selected from a dictionary to form global embeddings (Kohli et al., 2021). The method is particularly relevant since it constructs such isometries through greedy search, with putative dictionary features added one at a time.

That $D\phi$ is orthonormal has several equivalent formulations. We introduce a ground truth loss that is motivated by the following.

Proposition 1 *The singular values $\sigma_1 \dots \sigma_D$ are equal to 1 if and only if $U \in \mathbb{R}^{D \times D}$ is orthonormal.*

On the other hand, the formulation that motivates our convex approach is that orthonormal matrices consist of D coordinate features whose gradients are orthogonal and of unit length.

Proposition 2 *The component vectors $u_1 \dots u_D \in \mathbb{R}^B$ form a orthonormal matrix if and only if, for all $d_1, d_2 \in [D]$, $\langle u_{d_1}, u_{d_2} \rangle = \begin{cases} 1 & d_1 = d_2 \\ 0 & d_1 \neq d_2 \end{cases}$.*

2.3 Best subset selection

Given a matrix $X \in \mathbb{R}^{D\times P}$, we compare algorithmic paradigms for solving problems of the form

$$\arg \min_{S\in\binom{[P]}{D}} \ell(X_{.S}) \tag{3}$$

where $\binom{[P]}{D} = \{A \subseteq [P] : |A| = D\}$. Brute force algorithms consider all possible solutions. These algorithms are conceptually simple, but have the often prohibitive time complexity $O(C_\ell P^D)$ where C_ℓ is the cost of evaluating ℓ. Greedy algorithms consist of iteratively adding one element at a time to S. This algorithms have time complexity $O(C_\ell PD)$ and so are computationally more efficient than brute force algorithms, but can get stuck in local minima. Formal definitions are given in Section 6.1.

Sometimes, it is possible to introduce an objective which convexifies problems of the above form (Liberti, 2004; Abdi, 2013; Zhou and Low, 2021). Solutions

$$\arg\min f(\beta) : Y = X\beta \tag{4}$$

to the overcomplete regression problem $Y = X\beta$ are a classic example (Scott Shaobing Chen and David L. Donoho and Michael A. Saunders, 2001). When $f(\beta) = \|\beta\|_0$, this problem is non-convex, and is thus suitable for greedy or brute algorithms, but when $f(\beta) = \|\beta\|_1$, the problem is convex, and may be solved efficiently via interior-point methods. When the equality constraint is relaxed, Lagrangian duality may be used to reformulate as a so-called Lasso problem, which leads to an even richer set of optimization algorithms (Hastie et al., 2016).

The form of basis pursuit that we apply is inspired by the group basis pursuit (Koelle et al., 2022; Szeghy et al., 2022). Multitask basis pursuit is a special form of group basis pursuit which we call differently since grouping is dependent only on the structure of matrix-valued response variable y. The objective $f(\beta) = \|\beta\|_{1,2} := \sum_{p=1}^{P} \|\beta_{p.}\|_2$ creates joint sparsity across entire rows $\beta_{p.}$ (Yuan and Lin, 2006; Obozinski et al., 2006; Yeung and Zhang, 2011). It was used in Koelle et al. (2022) to select independent subsets among interpretable features.

3 Method

We apply the group lasso paradigm used to select independent dictionary elements in Koelle et al. (2022; 2024) to the more specific problem of selecting pointwise isometries from a dictionary. We first define a ground truth objective computable via brute and greedy algorithms that is uniquely minimized by orthonormal matrices. We then define the combination of normalization and multitask basis pursuit that approximates this ground truth loss function. We finally give a brute post-processing method for ensuring that the solution is D sparse, and provide a theoretical result that the two stage approach will always result in recovery of an isometric solution from the dictionary, should one exist.

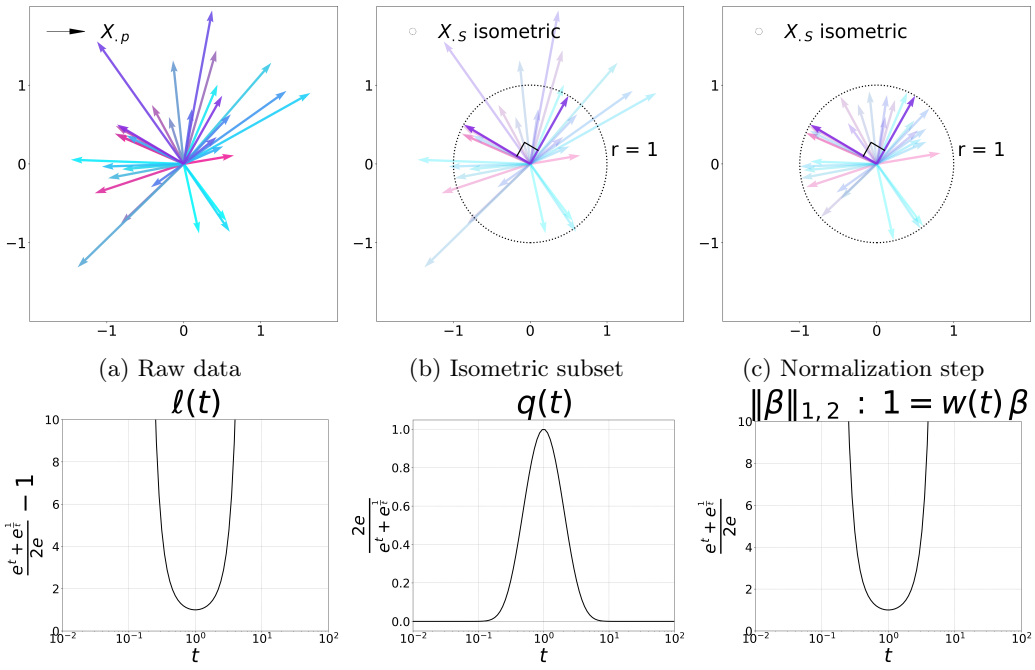
3.1 Ground truth

We'd like a ground truth objective to be minimized uniquely by orthonormal matrices, invariant under rotation, and depend on all changes in the matrix. Deformation (Kohli et al., 2021) and nuclear norm (Boyd and Vandenberghe, 2004) use only a subset of the differential's information and are not uniquely minimized at unitarity, respectively. We therefore introduce an alternative ground truth objective that satisfies the above desiderata and has convenient connections to isometry pursuit.

This ground truth objective is

$$\ell : \mathbb{R}^{D\times P} \to \mathbb{R}^+ \tag{5}$$

$$X \mapsto \sum_{d=1}^{D} g(\sigma_d(X)) \tag{6}$$



(a) Raw data (b) Isometric subset (c) Normalization step

(d) Ground truth loss (e) Normalized length vs. unnormalized length (f) Basis pursuit loss

Figure 1: 1a Raw data X represented as column-vectors. 1b An isometric subset - the identification of which is our objective. 1c The vectors after normalization so that vectors of length 1 maintain longest length. 1d Our ground truth loss function in the one-dimensional case $D = 1$ 1e Normalized length used to rescale Figure 1a to get vectors in Figure 1c. 1f Basis pursuit loss. This loss is equivalent to that shown in Figure 1d in the one-dimensional case.

where $\sigma_d(X)$ is the d-th singular value of X and

$$g : \mathbb{R}^+ \to \mathbb{R}^+ \tag{7}$$

$$t \mapsto \frac{e^t + e^{t^{-1}}}{2e}. \tag{8}$$

By Proposition 1, we can see that ℓ is uniquely maximized by orthonormal matrices. Moreover, g is convex, and $\ell(X^{-1}) = \ell(X)$ when X is invertible. Figure 1d gives a graph of ℓ when $D = 1$.

Our ground truth objective is therefore the best subset selection problem

$$\arg \min_{S \in \binom{[P]}{d}} \ell(X_{\cdot S}). \tag{9}$$

Regardless of the convexity of ℓ, brute combinatorial search over $[P]$ is inherently non-convex. This motivates our alternative formulation.

3.2 Normalization

Since basis pursuit methods tend to select longer vectors, selection of orthonormal submatrices requires normalization such that both long and short candidate basis vectors are penalized in the subsequent regression. We introduce the following definition.

Definition 3 (Symmetric normalization) *A function $q : \mathbb{R}^D \to \mathbb{R}^+$ is a symmetric normalization if*

$$\arg\max_{v\in\mathbb{R}^D} \; q(v) = \{v : \|v\|_2 = 1\} \tag{10}$$

$$q(v) = q(\frac{v}{\|v\|_2^2}) \tag{11}$$

$$q(v_1) = q(v_2) \; \forall \; v_1, v_2 \in \mathbb{R}^D : \|v_1\|_2 = \|v_2\|_2. \tag{12}$$

We use such functions to normalize column-vector length in such a way that vectors of length 1 prior to normalization have longest length after normalization and vectors are shrunk proportionately to their deviation from 1. That is, we normalize vectors by

$$n : \mathbb{R}^D \to \mathbb{R}^D \tag{13}$$

$$v \mapsto q(v)v \tag{14}$$

and matrices by

$$w : \mathbb{R}^{D\times P} \to \mathbb{R}^D \tag{15}$$

$$X_{\cdot p} \mapsto n(X_{\cdot p}) \; \forall \; p \in [P]. \tag{16}$$

We choose q as follows.

$$q : \mathbb{R}^D \to \mathbb{R}^+ \tag{17}$$

$$v \mapsto \frac{e^{\|v\|_2} + e^{\|v\|_2^{-1}}}{2e}. \tag{18}$$

Besides satisfying the conditions in Definition 3, this normalization has some additional nice properties. First, q is convex. Second, it grows asymptotically log-linearly. Third, while $\exp(-|\log t|) = \exp(-\max(t, 1/t))$ is a seemingly natural choice for normalization, it is non smooth, and the LogSumExp (Boyd and Vandenberghe, 2004) replacement of $\max(t, 1/t)$ with $\log(\exp(t)+\exp(1/t))$ simplifies to 18 upon exponentiation. This normalization provides good numerical properties for avoiding issues arising close to 0 and ∞. We plot this normalization function in Figure 1e and its effect on our example data in Figure 1c.

3.3 Isometry pursuit

Isometry pursuit is the application of multitask basis pursuit to the normalized design matrix $w(X)$ to identify submatrices of X that are as orthonormal as possible. Define the multitask basis pursuit penalty

$$\|\cdot\|_{1,2} : \mathbb{R}^{P\times D} \to \mathbb{R}^+ \tag{19}$$

$$\beta \mapsto \sum_{p=1}^{P} \|\beta_{p\cdot}\|_2. \tag{20}$$

Given a matrix $Y \in \mathbb{R}^{D\times D}$, the multitask basis pursuit solutions are

$$\widehat{\beta}_{MBP}(X, Y) := \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \; : \; Y = X\beta. \tag{21}$$

The multitask basis pursuit solution is often not unique. See Schneider and Tardivel (2022) for more details on uniqueness of basis pursuit solutions, for which the uniqueness condition is less strict than in the multitask case. We therefore define

$$\widehat{\beta}^{\ell_2}_{MLP}(X, Y) := \arg\min \|\beta\|_F : \beta \in \beta_{MBP}(X, Y). \tag{22}$$

By the strong convexity of the ℓ_2 norm, $\widehat{\beta}^{\ell_2}_{MLP}(X, Y)$ is well defined.

Isometry pursuit is then given by

$$\widehat{\beta}(X) := \widehat{\beta}^{\ell_2}_{MBP}(w(X), I_D) \tag{23}$$

where I_D is the D dimensional identity matrix and recovered functions are the indices of the dictionary elements with non-zero coefficients. That is, they are given by $S(\beta)$ where

$$S : \mathbb{R}^{P\times D} \to \binom{[P]}{D} \tag{24}$$

$$\beta \mapsto \{p \in [P] : \|\beta_{p\cdot}\| > 0\}. \tag{25}$$

IsometryPursuit(Matrix $X \in \mathbb{R}^{D\times P}$)

1: Normalize $X' = w(X)$
2: Optimize $\widehat{\beta} = \widehat{\beta}^{\ell_2}_{MBP}(X', I_D)$
3: **Output** $\widehat{S} = S(\widehat{\beta})$

3.4 Theory

The intuition behind our application of multitask basis pursuit is that submatrices consisting of vectors which are closer to 1 in length and more orthogonal will have smaller loss. A key theoretical assertion is that IsometryPursuit is invariant to choice of basis for X.

Proposition 3 *Let $U \in \mathbb{R}^{D\times D}$ be orthonormal. Then $S(\widehat{\beta}(UX)) = S(\widehat{\beta}(X))$.*

This proposition establishes the basis-independence of Isometry Pursuit and that we may replace I_D in the constraint by any orthonormal $D \times D$ matrix. It is also essential for showing the following main theoretical result.

When a rank D orthonormal column-submatrix $X_{\cdot S}$ exists, the output of Program 23 will contain S.

Proposition 4 *Let $X \in \mathbb{R}^{D\times P}$ have a rank D orthonormal column submatrix $X_{\cdot S}$. Then $S \subseteq \widehat{\beta}(X)$.*

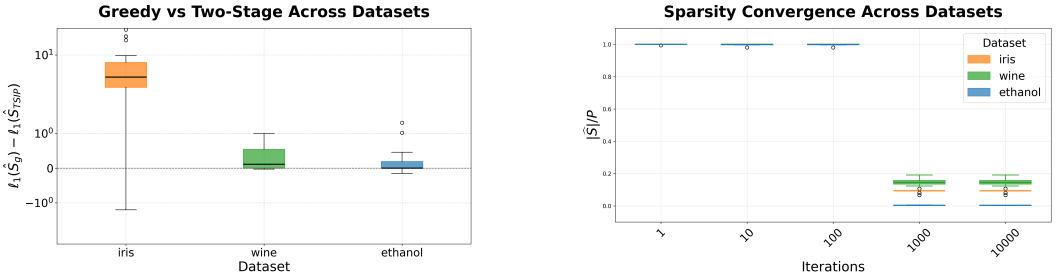
Proofs of these propositions are given in Section 6.2, with corresponding experimental results in Section 4. The proof mostly relies on the KKT conditions (Hastie et al., 2015) being used to verify that an isometric subset yields a corresponding basis pursuit solution for an appropriate normalized matrix. This is a formalization of our intuition that longer and more orthogonal vectors will minimize loss. We then apply a simple argument to show that the ℓ_2 minimizing solution is the least sparse solution, which means that it contains the isometry solution, should it exist.

3.5 Two-stage isometry pursuit

Proposition 4 suggests the following algorithm, which first uses the convex IsometryPursuit algorithm to prune the P candidate features and then applies brute search upon the reduced feature set. The proposition says that this approach is guaranteed to select an isometric submatrix should one exist. The two-stage approach is standard in the Lasso literature (Meinshausen, 2007; Hesterberg et al., 2008; Koelle et al., 2022), and Proposition 4 justifies it in our application. This method forms our practical isometry estimator, the core advantage of which over purely brute search is a substantially improved computational complexity.

TwoStageIsometryPursuit(Matrix $X \in \mathbb{R}^{D\times P}$)

1: $\widehat{S}_{IP} = \text{IsometryPursuit}(X)$
2: $\widehat{S} = \text{BruteSearch}(X_{\cdot \widehat{S}_{IP}}, \ell)$
3: **Output** $\widehat{S}$



(a) Loss improvement across datasets

(b) Sparsity convergence across datasets

Figure 2: Difference in isometry losses $\ell(\widehat{S}_g) - \ell(\widehat{S}_{TSIP})$ for Wine, Iris, and Ethanol datasets. The distributions reflect the relative losses of the supports estimated by greedy optimization on the ℓ objective is less than or equal to ℓ loss of the support estimated by TwoStageIsometryPursuit. As shown here and detailed in Table 1, losses are generally lower for two-stage isometry pursuit solutions.

3.6 Computation complexity

IsometryPursuit is more computationally efficient than BruteSearch search for the minimizer of ℓ. Interestingly, their asymptotic per-iteration complexities are the same.

Proposition 5 *The computational complexity C_ℓ of 5 is $O(D^2P)$.*

The above proposition follows immediately from the computational complexity of singular value decomposition (Liang et al., 2016). The per-iteration complexity of IsometryPursuit is also limited by the computational complexity of singular value decomposition, albeit in the more intricate manner described in Section 6.2.3.

Proposition 6 *The computational complexity of each step of Isometry Pursuit is $O(D^2P)$.*

The difference therefore lies in the relation of the number of iterations required to achieve a given error tolerance with IsometryPursuit and BruteSearch. To optimize, we use the SCS interior point solver (O'Donoghue et al., 2016) from CVXPY (Diamond and Boyd, 2016; Agrawal et al., 2018), which is able to push sparse values arbitrarily close to 0 (CVXPY Developers). SCS (aka ADMM) has asymptotic (in error) accuracy $\epsilon < O(\frac{1}{\mathcal{S}})$, where $\mathcal{S}$ is the number of iterations (He and Yuan, 2012). This means that the number of iterations required to achieve a given error tolerance scales inversely with the desired level of accuracy. We examine the effect of iteration number on the convergence to a sparse solution in Section 4.

4 Experiments

Say you are hosting a dinner party, and wish to select a balanced set of wines for drinking and flowers for decoration. We demonstrate TwoStageIsometryPursuit and GreedySearch on the Iris and Wine datasets (Fisher, 1988; Aeberhard and Forina, 1991; Pedregosa et al., 2011). This has an intuitive interpretation as selecting diverse elements that reflects the peculiar structure of the diversification problem. Features like *petal width* are rows in X. Thus, in diversification, $P = n$. Due to differences in scales between features, these are first standardized. For the Wine dataset, even BruteSearch on $\widehat{S}_{IP}$ is prohibitive in $D = 13$, and so we sample our inputs to $D = 6$.

Our main interpretability dataset is the Ethanol dataset from Chmiela et al. (2018); Koelle et al. (2022). For this dataset evaluate a dictionary of interpretable features - bond torsions - for their ability to parameterize the molecular configuration space. In this interpretability use case, columns denote gradients of informative features. We compute Jacoban matrices of putative parametrization functions and project them onto estimated tangent spaces (see Koelle et al. (2022) for preprocessing details). Rather than selecting between data points,

Name	D	P	R	$\ell(X_{\cdot\widehat{S}_G})$	$\lvert\widehat{S}\rvert$	$\ell(X_{\cdot\widehat{S}})$	$\widehat{P}(\ell(X_{\cdot\widehat{S}_G}) > \ell(X_{\cdot\widehat{S}}))$	$\widehat{P}(\ell(X_{\cdot\widehat{S}_G}) = \ell(X_{\cdot\widehat{S}}))$	$\widehat{P}(\bar{\ell}_1(X_{\cdot\widehat{S}_G}) > \bar{\ell}_1(X_{\cdot\widehat{S}}))$
Iris	4	75	25	14 ± 7.3	6.7 ± 1	6.9 ± 1.4	0.96	0	2.4e-05
Wine	6	89	25	7.7 ± 0.33	13 ± 1.5	7.6 ± 0.29	0.64	0.16	6.3e-04
Ethanol	2	756	25	2.6 ± 0.3	90 ± 1.7e+02	2.5 ± 0.2	0.66	0.17	2.1e-05

Table 1: Experimental parameters and results. For Iris and Wine, probability estimates $\widehat{P}$ result from random downsampling features $[P]$ by a factor of 2 to create R replicates, while for Ethanol, replicates correspond to individual data points. Distributional probabilities $\widehat{P}(\ell(X_{\cdot\widehat{S}_G}) > \ell(X_{\cdot\widehat{S}_{TSIP}}))$ and $\widehat{P}(\ell(X_{\cdot\widehat{S}_G}) = \ell(X_{\cdot\widehat{S}_{TSIP}}))$ are empirical across replicates, while asymptotic probabilities $\widehat{P}(\bar{\ell}_1(X_{\cdot\widehat{S}_G}) > \bar{\ell}_1(X_{\cdot\widehat{S}_{TSIP}}))$ is computed by paired two-sample T-test on $\ell(X_{\cdot\widehat{S}})$ and $\ell(X_{\cdot\widehat{S}_G})$. For brevity, in this table $\widehat{S} := \widehat{S}_{TSIP}$.

we are selecting between functions which parameterize the data. Replicas are created by sampling from data points and their corresponding tangent spaces are estimated in $B = 252$. This dataset exemplifies our motivating example - the search for locally isometric embeddings.

Figure 2 and Table 1 show that the ℓ accrued by the subset $\widehat{S}_G$ estimated using GreedySearch with objective ℓ is higher than that for the subset estimated by TwoStageIsometryPursuit. This effect is statistically significant across datasets. Generally, the ground truth loss obtained by GreedySearch optimization of ℓ is higher and therefore worse than that obtained by two-stage isometry pursuit. Table 1 and Figure 3 detail intermediate support recovery cardinalities from IsometryPursuit. Isometry pursuit substantially reduced the size of the dictionary and made brute search computationally feasible. Wall-clock runtimes are given in Section 6.4. These show that the preference for the two-stage isometry pursuit solution is strongest around $c = 1$ and is consistent across feature truncation dimensions.

5 Discussion

We have shown that multitask basis pursuit can help select isometric submatrices from appropriately normalized wide matrices. This approach - IsometryPursuit - is a convex alternative to greedy methods for selection of orthonormalized features from within a dictionary. Isometry pursuit can be applied to diversification and geometrically-faithful coordinate estimation. Our experiments exemplify these applications, but more can be done.

A principal drawback of IsometryPursuit is its reliance on the second stage of coordinate selection to complete the estimation process. D-sparse selection is sometimes computationally infeasible using BruteSearch (e.g. for Wine when $D = 13$). Moreover, we often may want to select some orthonormal subset of dimension much less than D. This is a particularly attractive idea for diversification in recommendation (Carbonell and Goldstein, 1998a; Wu et al., 2019; Lan) and retrieval (Gao et al., 2023; Pickett et al., 2024; In et al., 2024; Weiss, 2024; Vec). An avenue for solving this issue is the multitask Lasso (Hastie et al., 2015) extension of our estimator. Using the Lasso penalty, we can smoothly move between the multitask basis pursuit solution and more sparse solutions (**?**). This is particularly relevant as maximum cosine similarity used in retrieval corresponds to the first coefficient of the Lasso regularization path (Koelle et al., 2022). However, the standard max-sum diversity formulation that seeks to identify datapoints that are as far apart as possible is substantially different from the notion of diversity selected for by IsometryPursuit (Kuo et al., 1993; Ghosh, 1996; Alfonso Cevallos, Friedrich Eisenbrand, and Rico Zenklusen, 2016; Ashkan et al., 2015). More investigation is needed to compare against the substantial literature of greedy algorithms (Carbonell and Goldstein, 1998b; Barioni et al., 2011; Drosou and Pitoura, 2010; Qin et al., 2012; Kunaver and Požrl, 2017; Guo and Sanner, 2010; Abdool et al., 2020; Yu et al., 2016; Huang et al., 2024; Pickett et al., 2024; Cai and Wang, 2011; Zhu et al., 2020).

The comparison of greedy (Mallat and Zhang, 1993; 1992; Pati et al., 1993; Tropp et al., 2005) and convex (Scott Shaobing Chen and David L. Donoho and Michael A. Saunders,

2001; Tropp, 2006; Chen and Huo, 2006) basis pursuit formulations has a rich history, and theoretical understanding of the behavior of this approximation is evolving. For example, that the solution of a lasso problem can sometimes be a non-singleton set is well-known (Osborne et al., 2000; Dossal, 2012; Chrétien and Darses, 2011; Tibshirani, 2012; Ewald and Schneider, 2017; Ali and Tibshirani, 2018; Schneider and Tardivel, 2020; Mishkin and Pilanci, 2022; Dupuis and Vaiter, 2019; Debarre et al., 2020; Everink et al., 2024), but IsometryPursuit can generate non-unique solutions even when the design matrix satisfies general position condition. Besides the question of conditions for uniqueness, some other theoretical question posed by IsometryPursuit are how well the minimizer of a convex loss approximates the singular value loss, and the relationship with curvature.

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6 Supplement

This section contains algorithms, proofs, and experiments in support of the main text.

6.1 Algorithms

We give definitions of the brute and greedy algorithms for the combinatorial problem studied in this paper. The brute force algorithm is computationally intractable for all but the smallest problems, but always finds the global minima.

BruteSearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f)

for each combination $S \subseteq \{1, 2, \ldots, P\}$ with $|S| = D$ **do**
  Evaluate $f(X_{.S})$
end for
Output the combination S^* that minimizes $f(X_{.S})$

Greedy algorithms are computationally expedient but can get stuck in local optima (Cormen et al., 2009; Russell and Norvig, 2009), even with randomized restarts (Dick et al., 2014).

GreedySearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f, selected set $S = \emptyset$, current size $d = 0$)

if $d = D$ **then**
  Return S
else
  Initialize $S_{\text{best}} = S$
  Initialize $f_{\text{best}} = \infty$
  for each $p \in \{1, 2, \ldots, P\} \setminus S$ **do**
    Evaluate $f(X_{.(S \cup \{p\})})$
    if $f(X_{.(S \cup \{p\})}) < f_{\text{best}}$ **then**
      Update $S_{\text{best}} = S \cup \{p\}$
      Update $f_{\text{best}} = f(\mathcal{X}_{.(S \cup \{p\})})$
    end if
  end for
  Return GreedySearch(X, f, S_{best}, $d + 1$)
end if

6.2 Proofs

6.2.1 Proof of Proposition 3

In this proof we first show that the penalty $\|\beta\|_{1,2}$ is unchanged by unitary transformation of β.

Proposition 7 *Let $U \in \mathbb{R}^{D \times D}$ be unitary. Then $\|\beta\|_{1,2} = \|\beta U\|$.*

Proof:

$$\|\beta U\|_{1,2} = \sum_{p=1}^{P} \|\beta_{p.} U\| \tag{26}$$

$$= \sum_{p=1}^{P} \|\beta_{p.}\| \tag{27}$$

$$= \|\beta\|_{1,2} \tag{28}$$

□

We then show that this implies that the resultant loss is unchanged by unitary transformation of X.

Proposition 8 *Let $U \in \mathbb{R}^{D\times D}$ be unitary. Then $\widehat{\beta}(UX) = \widehat{\beta}(X)U$.*

Proof:

$$\widehat{\beta}(UX) = \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = UX\beta \tag{29}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ U^{-1}U = U^{-1}UX\beta U \tag{30}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = X\beta U \tag{31}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta U\|_{1,2} \ : \ I_D = X\beta U \tag{32}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = X\beta. \tag{33}$$

□

6.2.2 Proof of Proposition 4

In this proof we first show that the orthonormal submatrix $X_{\cdot S}$ is contained within the set $\widehat{\beta}_{MBP}(w(X), I_D)$.

Proposition 9 *Let $X_{\cdot S}$ be a rank D orthonormal submatrix of X. Then $S \in S(\beta) : \beta \in \widehat{\beta}_{MBP}(w(X), I_D))$.*

Proof: By Proposition 3 without loss of generality, let $X_{\cdot S} = I_D$. Also, without loss of generality let $S = \{1 \dots D\}$. We show that $\beta = [I_D \ 0]$ satisfies the KKT conditions for $X = [I_D \ X_{-S}]$.

Recall that the KKT conditions for a convex optimization problem are a set of conditions on an element of the domain of the optimization algorithm that, if satisfied, certify that the element is a solution. To write out the KKT conditions, we define the Lagrangian

$$\mathcal{L}(X, \beta, \nu) = \|\beta\|_{1,2} + \nu^T(I_D - w(X)\beta) \tag{34}$$

where $\nu \in \mathbb{R}^{D\times D}$ is the dual variable.

The KKT conditions are then

- Primal feasibility: $w(X)\beta = I_D$
- Stationarity: there exists a dual variable ν such that $0 \in \partial\|\beta\|_{1,2} - w(X)^T\nu$ where ∂ is the subdifferential operator.

where dual feasibility and complementary slackness are ignorable by virtue of the absence of inequality constraints.

First note that in our case, primal feasibility is satisfied by X being rank D and w not effecting the rank of the transformed matrix since it only rescales length and not direction.

For stationarity, recall that

$$\|\beta\|_{1,2} = \sum_{p=1}^{P} \|\beta_{p\cdot}\|_2. \tag{35}$$

Then, by the definition of subdifferential

$$\partial\|\beta\|_{1,2} = \begin{bmatrix} I_D v_{d+1} \\ \cdots \\ v_P \end{bmatrix} \tag{36}$$

where $v_p \in \mathbb{R}^D$ can be any vector satisfying $\|v_p\|_2 \leq 1$.

We therefore must show that there exists a ν that satisfies

$$\begin{bmatrix} I_D \\ w(X_{-S})^T \end{bmatrix} \nu = \begin{bmatrix} I_D \\ V_{-S} \end{bmatrix}. \tag{37}$$

At this point we can see that in fact I_D is an appropriate choice of ν since normalization by w leads to vectors satisfying $\|w(X)\|_2 \leq 1$.

Since for this β, $S(\beta) = S$, we have shown that S is the support of a solution to $\beta_{MBP}(w(X), I_D)$. □

The next part of the proof handles the presence of non-unique solutions.

Proposition 10 $S(\widehat{\beta}(X)) = \cup\, S(\beta) \;:\; \beta \in \beta_{MBP}(w(X), I_D)$

Proof: The proposition follows from geometric properties of the solution set. We ignore the case for singleton solutions, for which the proposition is self-evident. Recall that $\widehat{\beta}(X) := \widehat{\beta}^{\ell_2}_{MBP}(w(X), I_D) = \arg\min_{\beta \in \mathbb{R}^{P\times D}} \|\beta\|_F : \beta \in \widehat{\beta}_{MBP}(X, I_D)$. By definition of $\widehat{\beta}_{MBP}(X, I_D)$, $\|\beta_{p.}\|_2 : \beta \in \widehat{\beta}_{MBP}(X, I_D)$ is an affine linear subspace of $\mathbb{R}^P$. The projection of the vector of minimum ℓ_2 distance from the origin to a affine space intersects that space perpendicularly. However, for all $\|\beta_{.p}\|_2$ such that β is on the intersection of this affine space with the coordinate hyperplanes formed by fixing any choices of $\|\beta_{.p}\|_2 = 0$, this vector doesn't intersect this space perpendicularly. Thus, $\widehat{\beta}(X)$ is not on the intersection of the solution polytope $\widehat{\beta}_{MBP}(w(X), I_D)$ with the coordinate hyperplanes. It is therefore maximally non-sparse with respect to any of the minimizing solutions. □

6.2.3 Proof of Proposition 6

Proof: The augmented split Lagrangian minimized by ADMM is

$$\mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z) = \|z\|_{1,2} + \mathrm{tr}(\nu^T(I_D - w(X)\beta)) + \rho\|I_D - w(X)\beta\|_F^2 + \gamma\|\beta - z\|_F^2 \tag{38}$$

This minimization problem is

$$\arg \min_{\beta, z \in P\times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z) \tag{39}$$

For a more detailed explanation of the reasons to prefer optimizing the split augmented Lagrangian over direct optimization of Equation 19 see Boyd (2010).

The three main subroutines (43 - 47) called by ADMM determine the per iteration complexity. The update rules of the ADMM algorithm used to optimize the augmented split Lagrangian are

$$\beta^{k+1} = \arg \min_{\beta \in P\times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z^k) \tag{40}$$

$$z^{k+1} = \arg \min_{z \in P\times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta^{k+1}, z) \tag{41}$$

$$\nu^{k+1} = \nu^k + \rho(I_D - w(X)\beta^{k+1}) \tag{42}$$

We will analyze the computational complexity of each of these steps by rewriting the update rules according to the specifics of the isometry pursuit problem.

The beta update Equation 43 projects the solution onto the feasible set.

$$\beta^{k+1} = \arg \min_{\beta \in P\times D} \mathrm{tr}({\nu^k}^T(I_D - w(X)\beta)) + \rho\|I_D - w(X)\beta\|_2^2 + \gamma\|\beta - z^k\|_2^2 \tag{43}$$

$$= (\rho\, w(X)^T w(X) + \gamma I_D)^{-1}(\rho\, w(X)^T I_D + \gamma z^k + w(X)^T \nu^k) \tag{44}$$

The computational complexity of this computation is dominated by the inverse - that is, $O(D^2P)$.

Subsequently, the z update Equation 45 is soft-thresholding operation onto a contracted level set $\|\|\|_{1,2}$ ball.

$$z^{k+1} = \arg\min_{z \in P \times D} \|z\|_{1,2} + \gamma \|z - \beta^{k+1}\|_2^2 \tag{45}$$

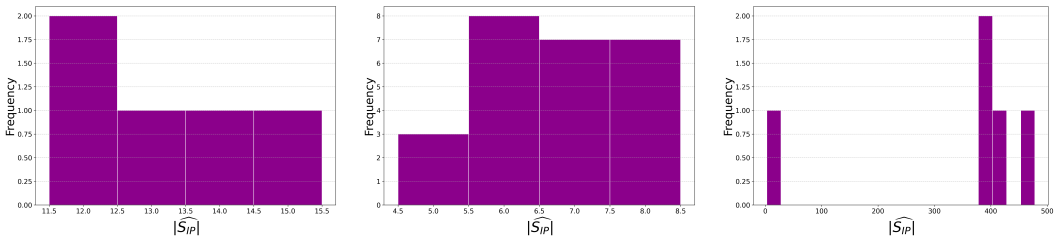
$$= \begin{cases} (1 - \frac{1}{\gamma \|\beta_{j.}^{k+1}\|_2})(\beta_{j.}^{k+1}) \text{ if } \|\beta_{j.}^{k+1}\|_2 > \frac{1}{\gamma} \\ 0 \text{ otherwise} \end{cases} \tag{46}$$

The computational complexity of this operation is $O(PD)$.

Finally, the ν update Equation 47 increases the squeezing of the z and β variables towards to each other.

$$\nu^{k+1} = \nu^k + \rho(I_D - w(X)\beta^{k+1}) \tag{47}$$

In this step, the dual variable ν that controls the penalty accrued by the splitting is updated in the direction of the constraint violation. The computational complexity of this operation is dominated by that of the matrix multiplication $w(X)\beta^{k+1}$. That is, it is $O(D^2P)$. □



(a) Wine (b) Iris (c) Ethanol

Figure 3: Support cardinalities for each dataset after IsometryPursuit.

6.3 Support cardinalities

Figure 3 plots the distribution of $|\widehat{S}_{IP}|$ from Table 1 in order to contextualize the reported means. While typically $|\widehat{S}_{IP}| << P$, there are cases for Ethanol where this is not the case that drive up the means. Further investigation into the extreme failure of isometry pursuit to sparsify in the the minor chunk of the bimodal Ethanol support cardinality distribution is warranted.

6.4 Timing

While wall-time of algorithms is a non-theoretical quantity that depends on implementation details, it provides valuable context for practitioners. We therefore report the following runtimes on a 2021 Macbook Pro. The particularly high variance for brute force search in the second step of TwoStageIsometryPursuit is likely due to the large cardinalities reported in Figure 3.

Name	IP	2nd stage	Greedy
Iris	1.24 ± 0.02	0.00 ± 0.00	0.02 ± 0.00
Wine	2.32 ± 0.17	0.13 ± 0.12	0.03 ± 0.00
Ethanol	8.38 ± 0.57	0.55 ± 1.08	0.07 ± 0.01

Table 2: Algorithm runtimes in seconds across replicates.